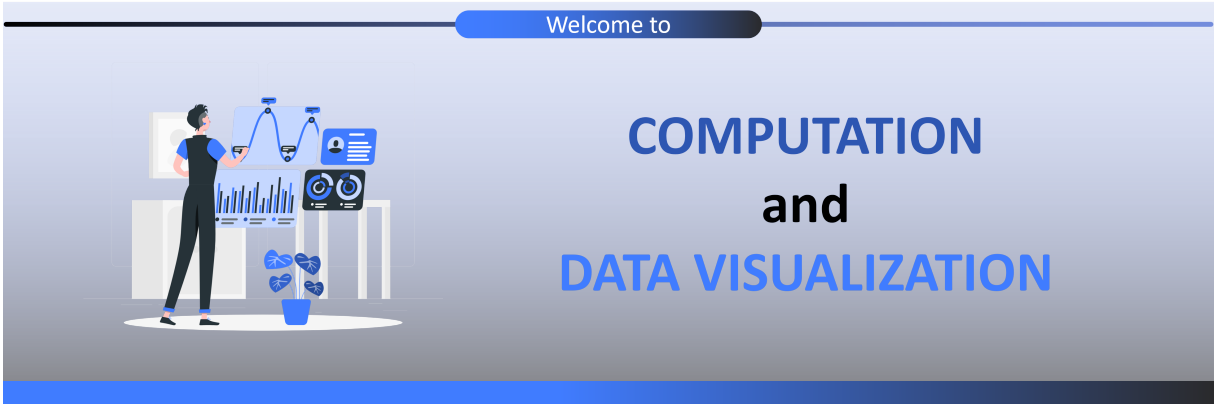
IE 6600: Computation and Data Visualization



Prof. Mohammad Dehghani

"The mind is not a vessel to be filled but a fire to be kindled"
- Plutarch, Greek philosopher

IE6600

Computation and Data Visualization

Instructor Info

Mohammad Dehghani

Office Hrs:

TBD

Website

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Course Info

None

Weekly Sessions: As per the itinerary

Time: As mentioned in schedule

TA Info

Check Canvas

Check Canvas

In-class

Check Canvas

Course Overview

Offers students an opportunity to learn how to use visualization tools and techniques for data exploration, knowledge discovery, data storytelling, and decision-making in engineering, healthcare operations, manufacturing, and related applications. It covers the basics of Python and R for data mining and visualization and introduces both static and interactive techniques that reveal insights, patterns, interactions, and comparisons by focusing on key elements such as color encoding, shape selection, spatial layout, and annotation. The core values of the course are: i) Data Analytics â enabling students to analyze and extract meaningful insights from complex datasets, ii) Data Visualization â teaching students to effectively represent data using visual techniques, and iii) Narrativeâ developing storytelling and presentation skills to convey data-driven insights clearly and persuasively.

- Data analysis: Students will learn how to analyze and clean data using various tools and techniques to make data-driven decisions. This includes data wrangling and exploratory data analysis.
- Visualization: Students will learn how to develop effective presentation skills and storytelling techniques to convey insights to diverse audiences. Additionally, students will understand the significance of storytelling in data presentation and craft a compelling message by structuring a narrative.
- Presentation and storytelling: Students will develop presentation skills, including slide creation, communication techniques, storytelling, narrative structure, and crafting a compelling message for diverse audiences.

Course Objectives

- Learn techniques for analyzing, cleaning, and summarizing data for making data-driven decisions
- Gain proficiency in data visualization, including static, dynamic, and interactive charts
- Develop skills in creating dashboards and applying business intelligence techniques for effective decision-making
- Understand the principles of data storytelling and how to present insights persuasively
- Explore the role of Generative AI in enhancing data visualization and storytelling
- Apply best practices in design and layout to develop compelling visualizations and confidently present data-driven stories to peers and stakeholders

Reference Books

Optional Textbooks:

1. *Storytelling with Data: A Data Visualization Guide for Business Professionals* by Cole Nussbaumer Knaflic (link)
2. *Storytelling with You: Plan, Create, and Deliver a Stellar Presentation* by Cole Nussbaumer Knaflic (link)

Outline

Highlights of some of the course's tentative topics and activities:

Major Topics

- Introduction to Storytelling with Data
- Data Cleaning and Data Wrangling Techniques
- Best Practices for Creating Effective Visualizations
- Dynamic and Interactive Charts
- The Role of Generative AI in Data Analysis and Visualization
- Tips and Tricks for Creating Effective Dashboards
- Essential Presentation Skills
- Creating a Compelling Story with Data

Software

- Coding: Python, R
- Interactive Charts: Plotly Library, Flourish Studio, Datawrapper
- Dashboards: Tableau, Power BI
- Generative AI Tools: Enhancing Data Storytelling and Visualization

Class Schedule

As shown in Figure 1, the course is designed in three phases, with each phase focusing on developing different skills in students, as illustrated in the figure below.

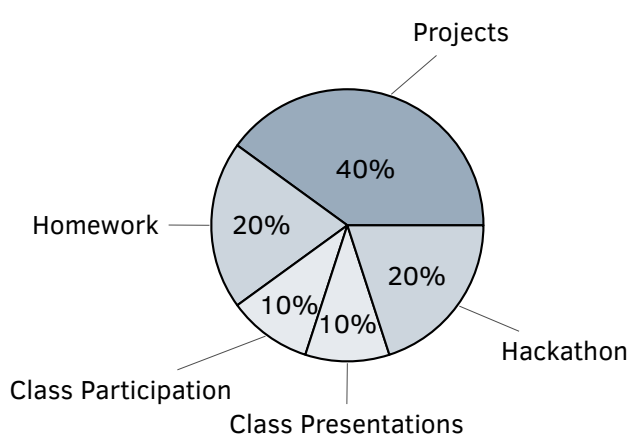
Figure 1: Course Phases: Data Processing, Interactive Storytelling, and Business Intelligence Dashboards

The course is structured into the following key phases:

- Phase 1: Data Processing & Visualization
  - Data wrangling and cleaning using R.
  - Exploratory data analysis (EDA) and pattern detection.
  - Data visualization using ggplot2 and base R plotting.
  - Project: Hackathon.
- Phase 2: Interactive Charts & Storytelling
  - Creating interactive charts using Flourish, Datawrapper, and Plotly.
  - Choosing effective visualization techniques for data narratives.
  - Project: Data storytelling website.
- Phase 3: Business Intelligence Dashboards
  - Dashboard design principles and best practices.
  - Hands-on practice with Power BI and Tableau.
  - Project: Business intelligence dashboards.

Please check CANVAS and the course itinerary for the detailed schedule of the course. If due to unforeseen circumstances the schedule changes, students will be informed.

Grading Scheme



- Homeworks: focus on improving the technical skills of the students. The homeworks will help the students gain expertise

in using Excel to perform basic data-wrangling operations and create visualizations.

- Hackathon: The hackathon will provide students with an opportunity to work in teams on a real-world data challenge. The goal is to apply the skills learned throughout the course to solve a complex problem within a limited time frame. Students will be evaluated based on creativity, technical implementation, and presentation of their solution.

- Project 1- Storytelling with Data: Students are expected to select a topic of interest and collect the relevant data. The insights from the data must be presented in form of visualizations. The submission would be a *GoogleSites* web page.

- Project 2- Dashboard Design: Students are expected to select a topic of interest and collect the relevant data. The dashboard should contain visual elements that allow the users to interact and gain insights from the data. The submission would be a *PowerBi dashboard* file.

- Class Participation: is divided into two components, 1) Attendance, and 2) Interaction during class, office hours, and interaction with the TA's. The course is synchronous and attendance is mandatory in person for students taking the course. We allow 2 absences that can be waived off and every missed class thereafter results in a deduction of 2 points. In case of an emergency, the student must email the professor and the TAs.

- Class Presentation:: Students will present a relevant topic related to the course. The topics and presentation references will be provided to students.

Your grades will be posted on CANVAS. You can check your running grade at any time, so there are no surprises. No grade corrections will be made after the last week of the semester. Final Grades and Course Letter Grades are not curved and are assigned as follows:

Letter Grade	Points	Letter Grade	Points	Letter Grade	Points
A	93-100	B	83-86	C	73-76
A-	90-92	B-	80-82	C-	70-72
B+	87-89	C+	77-79	F	<70

Important Teamwork Skills

This course has a lot of submissions that need you to work in a team. Therefore, it is essential for students to have a fruitful collaboration and effective communication with their teammates in order to succeed in this course. We encourage you to read this article which highlights 7 important teamwork skills.

Emailing Us?

Using a correct subject line helps us to respond your queries efficiently and timely. So, please use the following format for the subject line of your email to reach out to the professor or your TA. The subject line and an example are given below:

Course ID - Subject of Email:
IE6600: A short description of your query

Make-up Policy

- No makeup opportunity will be given for any homeworks/lab assignment/projects.
- Students unable to attend classes are responsible for obtaining information regarding announcements and lecture content

Class Policy

- Students are expected to attend all class sessions and are responsible for all materials presented in class. Past experience has shown students who attend regularly and interact with instructors/classmates tend to be better overall.
- Laptops: You may need to use your laptop during the class upon the instructor's notes
- Phones/tablets/smart watches: Electronic devices may not be used during the class. If you have class work, social networking or sports-score browsing, that is more important to do, then please skip class. Life is full of such choices.

- Discussion between students for homeworks and projects is permissible. However the ideas / solutions / codes must not be copied. Each individual submission must be an outcome of individual effort

Academic Integrity

A commitment to the principles of academic integrity is essential to the mission of Northeastern University. The promotion of independent and original scholarship ensures that students derive the most from their educational experience and their pursuit of knowledge. Academic dishonesty violates the most fundamental values of an intellectual community and undermines the achievements of the entire University.

As members of the academic community, students must become familiar with their rights and responsibilities. In each course, they are responsible for knowing the requirements and restrictions regarding research and writing, examinations of whatever kind, collaborative work, the use of study aids, the appropriateness of assistance, and other issues. Students are responsible for learning the conventions of documentation and acknowledgment of sources in their fields. Northeastern University expects students to complete all examinations, tests, papers, creative projects, and assignments of any kind according to the highest ethical standards, as set forth either explicitly or implicitly in this Code or by the direction of instructors.

Go to Academic Integrity website to access the full academic integrity policy.

Student accommodations

Northeastern University and the Disability Resource Center (DRC) are committed to providing disability services that enable students who qualify under Section 504 of the Rehabilitation Act and the Americans with Disabilities Act Amendments Act (ADAAA) to participate fully in the activities of the university. To receive accommodations through the DRC, students must provide appropriate documentation that demonstrates a current substantially limiting disability.

For more information, visit Disability Resource Center website

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Diversity and Inclusivity Statement

Northeastern University is committed to equal opportunity, affirmative action, diversity and social justice while building a climate of inclusion on and beyond campus. In the classroom, members of the University community work to cultivate an inclusive environment that denounces discrimination through innovation, collaboration and an awareness of global perspectives on social justice. It is my intention that students from all backgrounds and perspectives will be well served by this course, and that the diversity that students bring to this class will be viewed as an asset. I welcome individuals of all ages, backgrounds, beliefs, ethnicities, genders, gender identities, gender expressions, national origins, religious affiliations, sexual orientations, socioeconomic background, family education level, ability - and other visible and nonvisible differences. All members of this class are expected to contribute to a respectful, welcoming and inclusive environment for every other member of the class. Your suggestions are encouraged and appreciated.

Please visit the website for complete information on Diversity and Inclusion.

TITLE IX

Title IX of the Education Amendments of 1972 protects individuals from sex or gender-based discrimination, including discrimination based on gender-identity, in educational programs and activities that receive federal financial assistance.
Northeastern's Title IX Policy prohibits Prohibited Offenses, which are defined as sexual harassment, sexual assault, relationship or domestic violence, and stalking. The Title IX Policy applies to the entire community, including male, female, transgender students, faculty and staff.
In case of an emergency, please call 911.

Please visit the website for a complete list of reporting options and resources both on- and off-campus